

▼ Figure 33.8

MAKE CONNECTIONS

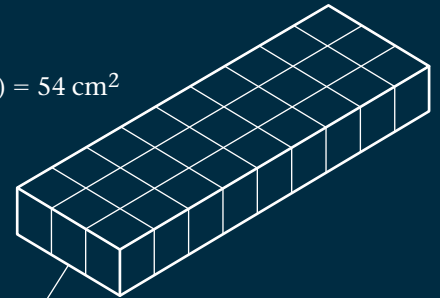
Maximizing Surface Area

In general, the amount of metabolic or chemical activity a cell or organism can carry out is proportional to its mass or volume. Maximizing metabolic rate, however, requires the efficient uptake of energy and raw materials, such as nutrients and oxygen, as well as the effective disposal of waste products. For large cells, plants, and animals, these exchange processes have the potential to be limiting due to simple geometry. When a cell or organism grows without changing shape, its volume increases more rapidly than its surface area (see Figure 6.7). As a result, there is proportionately less surface area over which exchange processes can occur. The challenge posed by the relationship between surface area and volume occurs in diverse contexts and organisms, but the evolutionary adaptations that meet this challenge are similar. Structures that maximize surface area through branching, flattening, folding, and projections have an essential role in biological systems.



$$SA: 6 (3 \text{ cm} \times 3 \text{ cm}) = 54 \text{ cm}^2$$

$$V: 3 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm} = 27 \text{ cm}^3$$



$$SA: 2 (3 \text{ cm} \times 1 \text{ cm}) + 2 (9 \text{ cm} \times 1 \text{ cm}) + 2 (3 \text{ cm} \times 9 \text{ cm}) = 78 \text{ cm}^2$$

$$V: 1 \text{ cm} \times 3 \text{ cm} \times 9 \text{ cm} = 27 \text{ cm}^3$$

These diagrams compare surface area (SA) for two different shapes with the same volume (V). Note which shape has the greater surface area.

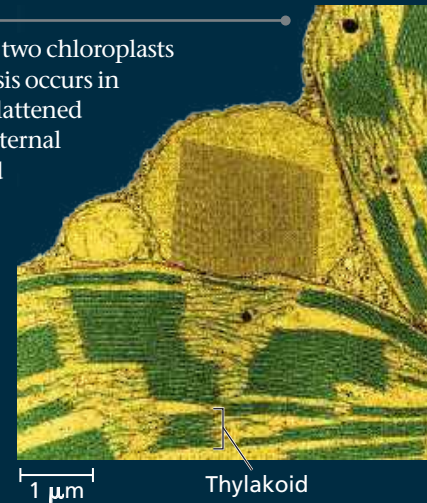
Branching

Water uptake relies on passive diffusion. The highly branched filaments of a fungal mycelium increase the surface area across which water and minerals can be absorbed from the environment. (See Figure 31.2.)



Folding

This TEM shows portions of two chloroplasts in a plant leaf. Photosynthesis occurs in chloroplasts, which have a flattened and interconnected set of internal membranes called thylakoid membranes. The foldings of the thylakoid membranes increase their surface area, enhancing the exposure to light and thus increasing the rate of photosynthesis. (See Figure 10.4.)



Flattening

By having a body that is only a few cells thick, an organism such as this flatworm can use its entire body surface for exchange. (See Figure 40.3)



Projections

In vertebrates, the small intestine is lined with finger-like projections called villi that absorb nutrients released by the digestion of food. Each of the villi shown here is covered with large numbers of microscopic projections called microvilli, resulting in a total surface area of about 300 m² in a human, as large as a tennis court. (See Figure 41.12.)



MAKE CONNECTIONS

Find other examples of flattening, folding, branching, and projections (see Chapters 6, 9, 35, and 42). How is maximizing surface area important to the structure's function in each example?